

Leveraging a Low-Cost RISC-V Microcontroller Board as Wearable Ultrasound Front End

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Motivation and Objective

Mobile and continuous health monitoring through wearable devices is increasingly used for medical diagnostic. Ultrasound (US) is gaining interest in this context because, unlike common sensors, it enables richer physiological assessment by adding biomechanical information, e.g., blood pressure. However, wearable devices put strict constraints on weight, size, thermal behaviour, and power consumption. Hence, the hardware for US transducer excitation and echo acquisition must combine low complexity, minimal form factor, and energy efficiency. This work demonstrates the feasibility of leveraging a \$5.00 commercial RISC-V microcontroller (μ C) board as an US front-end for the excitation signals generation.

Statement of Contribution

A prototype device was implemented by interfacing a two-pole cable with a Raspberry Pi Pico 2 board (panel A). Each pole was connected to 4 general purpose input/output (GPIO) pins, short-circuited together to form a single output node. The on-board RISC-V processor is programmed to configure the μ C as a current-driven nine-level digital to analog converter (DAC). The output amplitude is controlled by setting the two groups of GPIOs alternatively as input or output, and by enabling a number of outputs so that activating more pins increases the source current.

To enhance flexibility, the C code for the μ C programming is generated by a Python class: the tool synthesizes basic waveforms, e.g., sine waves and binary codes, and arbitrary complex signals through modulation, e.g., PSK modulation and maximum length sequences.

To test the prototype, the μ C was connected to a setup comprising a transmitter and a receiver transducers immersed in water facing a reflective metal plate, and an oscilloscope.

Results

Preliminary tests show that the prototype can generate sine waves and arbitrary PSK-modulated signals (panel B) with a 4V peak-to-peak amplitude. When the transmitter is driven with a 5-cycle sine burst, the receiver successfully detects the reflected echoes from the target (panel C).

In conclusion, the proposed DAC architecture yields nine discrete output levels, enabling the Raspberry Pi Pico 2 board to operate as a simple, low-cost excitation front-end able to synthesize arbitrary excitation. These results support its potential use in wearable US applications.

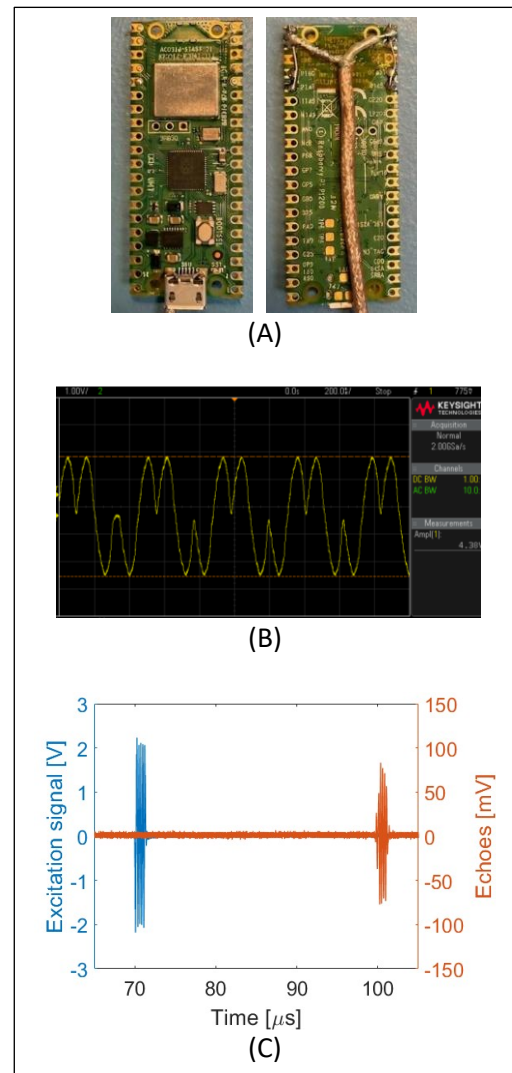


Figure. Prototype and results.